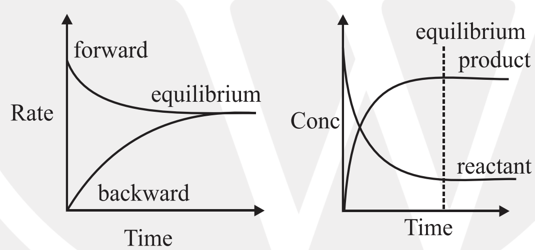
CHAPTER 7

Equilibrium

Sr. No.	Concept	Formulas	Other information
1.	**Condition of equilibrium**	(i) $R_f = R_b$ (ii) $\Delta G = 0$ (iii) $Q = K_{eq}$.	R_f = Rate of forward reaction R_b = Rate of backward reaction ΔG = Change in Gibbs free energy Q = Reaction quotient K_{eq} = Equilibrium constant
2.	**Graphical representation of equilibrium**		
3.	**Law of chemical equilibrium & equilibrium constant (K_c)**	For a reversible reaction $aA + bB \rightleftharpoons cC + dD$ $K_C = \frac{[C]^c[D]^d}{[A]^a[B]^b}$ $K_C = \frac{k_f}{k_b}$	K_C = Equilibrium constant k_f = Rate constant of forward reaction k_b = Rate constant of backward reaction
	(a) Equilibrium constant in terms of partial pressure (K_P)	$K_P = \frac{[P_C]^c\,[P_D]^d}{[P_A]^a\,[P_B]^b}$	P = Partial pressure
	(b) Equilibrium constant in terms of mole fraction (K_X)	$K_X = \frac{X_C^c X_D^d}{X_A^a X_B^b}$	X = Mole fraction

Sr. No.	Concept	Formulas	Other information
4.	**Relation between K_P & K_C**	$K_p = K_c(RT)^{\Delta n_g}$ (i) When $\Delta n_g = 0$ then $K_P = K_C$ (ii) When $\Delta n_g > 0$ then $K_P > K_C$ (iii) When $\Delta n_g < 0$ then $K_P < K_C$	K_c = Equilibrium constant Δn_g = change in no. of gaseous moles of products and reactants
5.	**Relation between equilibrium constant & standard Gibbs free energy change**	$\Delta G = -RT \ln K$ $\Delta G = -2.303RT \log K$ $K = e^{-\Delta G/RT}$ If $\Delta G < 0$ then $K > 1$ [Forward reaction is favoured.] If $\Delta G > 0$ then $K < 1$ [Reverse reaction is favoured.] If $\Delta G = 0$ then $K = 1$ [Reaction is in equilibrium.]	K = Equilibrium constant R = Gas constant T = Temperature ΔG = Gibbs free energy change
6.	**Reaction Quotient (Q_c)**	If the reaction is not at equilibrium, then 'Q_c' reaction quotient is expressed instead of 'K_c'. For reaction $aA + bB \rightleftharpoons cC + dD$ $Q_c = \frac{[C]^c[D]^d}{[A]^a[B]^b}$ If $Q_c = K_c$, the reaction is at equilibrium. If $Q_c > K_c$, the reaction will proceed in backward direction to decrease the value of Q_c. If $Q_c < K_c$, the reaction will proceed in forward direction to increase the value of Q_c.	Q_c = Reaction quotient K_c = Equilibrium constant
7.	**Degree of Dissociation (α)**	$\alpha = \frac{\text{no. of moles dissociated}}{\text{initial no. of moles taken}}$	
8.	**Le Chatelier's Principle** (a) Effect of concentration	Conc. of reactant increases $\longrightarrow$ shift towards forward reaction. Conc. of product increases $\longrightarrow$ shift towards backward reaction	
	(b) Effect of volume	If volume is increased, then (i) $\Delta n_g > 0$ reaction will shift in the forward direction (ii) $\Delta n_g < 0$ reaction will shift in the backward direction (iii) $\Delta n_g = 0$ reaction will not shift	Δn_g = change in no. of gaseous moles of products and reactants

Sr. No.	Concept	Formulas	Other information
	(c) Effect of pressure	Pressure increases $\longrightarrow$ shift towards lesser number of gaseous moles Pressure decreases $\longrightarrow$ shift towards higher number of gaseous moles	
	(d) Effect of inert gas addition at constant pressure	(i) $\Delta n_g > 0$ reaction will shift in the forward direction (ii) $\Delta n_g < 0$ reaction will shift in the backward direction (iii) $\Delta n_g = 0$ reaction will not shift	
	(e) Effect of Temperature	Addition of inert gas at constant volume $\rightarrow$ no effect. (i) Exothermic T increases $\rightarrow$ Reaction will shift in Backward direction (ii) Endothermic T increases $\rightarrow$ Reaction will shift in Forward direction.	Δn_g = change in no. of gaseous moles of products and reactants
9.	**Ostwald Dilution Law** (a) Dissociation constant	$\alpha \propto \sqrt{\text{dilution}}$, dilution $\uparrow \Rightarrow \alpha \uparrow$ $= \sqrt{\frac{K_a}{C}}$ (i) Weak acid $K_a = C\alpha^2$ (ii) Weak base $\alpha = \sqrt{\frac{K_b}{C}}$	K_a = Ionisation constant of acid K_b = Ionisation constant of base C = Concentration V = Volume α = Degree of dissociation
10.	**Acidity and pH scale**	$pH = -\log [H^+]$; $[H^+] = 10^{-pH}$ $pOH = -\log [OH^-]$; $[OH^-] = 10^{-pOH}$ $pk_a = -\log K_a$; $pK_b = -\log K_b$;	
11.	**Ionic product of water**	K_a ; $K_b = [H^+] [OH^-] = K_w = 10^{-14}$. $pK_a + pK_b = pK_w = 14$ at 25°C.	K_w = Ionic product of water.

Sr. No.	Concept	Formulas	Other information
12.	**pH calculations of different types of solutions**		
	(a) Strong acid solution:	(i) Concentration > 10^{-6} M	H^+ ions coming from water can be neglected
		(ii) Concentration < 10^{-6} M	H^+ ions coming from water cannot be neglected
	(b) pH of mixture of two strong acids	$[H^+] = N = \dfrac{N_1V_1 + N_2V_2}{V_1 + V_2}$	Number of H^+ ions from I-solution = N_1V_1 N = Normality Number of H^+ ions from II-solution = N_2 V_2
	(c) pH of mixture of two strong bases	$[OH^-] = N = \dfrac{N_1V_1 + N_2V_2}{V_1 + V_2}$	N = Normality
	(d) pH of mixture of a strong acid and a strong base	(i) If $N_1V_1 > N_2V_2$ solution will be acidic in nature $[H^+] = N = \dfrac{N_1V_1 - N_2V_2}{V_1 + V_2}$ (ii) If $N_2V_2 > N_1V_1$ solution will be basic in nature $[OH^-] = N = \dfrac{N_2V_2 - N_1V_1}{V_1 + V_2}$	N_1 = Normality of acidic component N_2 = Normality of basic component
	(e) pH of a weak acid(monoprotic) solution	$K_a = \dfrac{[H^+][OH^-]}{[HA]} = \dfrac{C\alpha^2}{1-\alpha}$ if $\alpha << 1 \Rightarrow (1-\alpha) \approx 1 \Rightarrow K_a \approx C\alpha^2$ $\alpha = \sqrt{\dfrac{K_a}{C}}$ (is valid if $\alpha < 0.1$ or 10%) pH = –log (Cα) On increasing the dilution $C\downarrow \Rightarrow \alpha\uparrow$ and $[H^+]\downarrow \Rightarrow pH\uparrow$	K_a = Ionisation constant of acid α = Degree of dissociation C = Concentration
13.	**Relative Strength of Two Acids**	$\dfrac{[H^+] \text{ furnished by I acid}}{[H^+] \text{ furnished by II acid}} = \dfrac{C_1\alpha_1}{C_2\alpha_2} = \sqrt{\dfrac{k_{a_1}C_1}{k_{a_2}C_2}}$	

Sr. No.	Concept	Formulas	Other information
14.	**Salt Hydrolysis**		
	(a) Strong acid & strong base	Do not hydrolysed, pH = 7	Neutral solution
	(b) Strong acid & weak base	$K_h = \frac{K_w}{K_b}$ $h = \sqrt{\frac{K_w}{K_b \times C}}$ $[H^+] = C.h. = \sqrt{\frac{K_w . C}{K_b}}$ $pH = 7 - \frac{1}{2} pK_b - \frac{1}{2} \log C$	Acidic solution h = Degree of hydrolysis K_h = Hydrolysis constant K_b = Ionisation constant of base C = Concentration of salt K_w = Ionic product of water
	(c) Weak acid & strong base	$K_h = \frac{K_w}{K_a}$ $h = \sqrt{\frac{K_w}{K_a \times C}}$ $[OH^-] = C.h. = \sqrt{\frac{K_w . C}{K_a}}$ $pH = 7 + \frac{1}{2} pK_a + \frac{1}{2} \log C$	Basic solution h = Degree of hydrolysis K_h = Hydrolysis constant K_a = Ionisation constant of acid C = Concentration of salt K_w = Ionic product of water
	(d) Weak acid & weak base	$K_h = \frac{K_w}{K_a \times K_b}$ $h = \sqrt{\frac{K_w}{K_a \times K_b}}$ $[H^+] = K_a . h = \sqrt{\frac{K_w . K_a}{K_b}}$ $[H^+] = \sqrt{\frac{K_w \times K_a}{K_b}}$ $pH = 7 + \frac{1}{2}[pk_a - pk_b]$	Almost neutral solution h = Degree of hydrolysis K_h = Hydrolysis constant K_a = Ionisation constant of acid K_b = Ionisation constant of base C = Concentration of salt K_w = Ionic product of water

Sr. No.	Concept	Formulas	Other information
15.	**Buffer Solution** (a) Acidic Buffer (b) Basic Buffer	[Henderson's equation] $pH = pK_a + \log\frac{[Salt]}{[Acid]}$ [Henderson's equation] $pOH = pK_b + \log\frac{[Salt]}{[Base]}$	$pK_a = -\log(K_a)$ $pK_b = -\log(K_b)$
16.	**Buffer capacity**	$\text{Buffer capacity} = \frac{\text{No. of moles acid or base added per litre of buffer}}{\text{Change in pH}}$	
17.	**Solubility Product**	$K_{SP} = (xs)^x (ys)^y = x^x.y^y.(s)^{x+y}$	K_{SP} = Solubility product
18.	**Condition For Precipitation**	(i) $K_{IP} < K_{SP}$ Unsaturated (ii) $K_{IP} > K_{SP}$ Precipitation (iii) $K_{IP} = K_{SP}$ Saturated	IP = Ionic product SP = Solubility product